

Millimetre-wave technologies.

Using frequencies much higher in the frequency spectrum opens up more spectrums and also provides the possibility of having much wide-channel bandwidth of possibly 1GHz-2GHz. However, this poses new challenges for handset development where maximum frequencies of around 2GHz and bandwidths of 10MHz-20MHz are currently in use.

For 5G, frequencies of above 50GHz are being considered. This will present some real challenges in terms of circuit design, technology and also the way the system is used, as these frequencies do not travel as far and are absorbed almost completely by obstacles.

Future PHY/MAC. This area presents many possibilities, from the use of new modulation formats including generalised frequency division multiplexing (GFD) as well as filter bank multi-carrier (FBMC), universal filtered multi-

carrier (UFMC) and other schemes to the management of multiple access schemes.

All these need to be developed. Higher levels of processing that will be available by the time 5G is launched means that multi-carrier systems will not require to be orthogonal as in the case of OFDM. This provides considerably more flexibility.

Massive MIMO. Although multiple-input multiple-output (MIMO) technology is being used in many applications from LTE to Wi-Fi, the number of antennae is fairly limited. Using microwave frequencies opens up the possibility of using many tens of antennae on a single equipment because of antenna size and spacing in terms of wavelength.

Dense networks. Reducing the size of cells provides a much more overall effective use of the available spectrum. Techniques to ensure that small cells in the macro-network

deployed as femto cells can operate satisfactorily are required.

Requirements of 5G network

From a network perspective, 5G requires tight and seamless interworking among existing and future standards. Rising demand for mobile traffic will enforce new ways of enhancing capacity, such as dense deployment of small cells, as well as intelligent traffic steering and off-load schemes.

Ever-growing energy consumption in wireless networks imposes new mechanisms of energy control and reduction. Finally, there is a need for autonomous network management because of network complexity and heterogeneity.

Devices are becoming more powerful and more numerous. Beyond devices like smartphones, tablets or game consoles, the future wireless landscape will have to serve cars, smart grid terminals, health-moni-



Would You Like More DIY Circuits?

We Have Thousands!

VISIT TODAY

electronicsforu.com

If it's electronics, it's here